Java Programming

Case Study 34. Blood Donation Camp Management System

Problem Statement

A healthcare organization conducts blood donation camps and requires a system to manage donors, donation appointments, blood groups, donation records, and certificates. The system should help staff track donor participation and blood collection.

Objectives

- Register blood donors.
- Manage donation appointments.
- Record blood donations.
- Search donors based on blood group.
- Maintain donation history.
- Generate donor reports and certificates.

Required Java Implementation

Java Concept	Application
Classes & Objects	Donor, Donation, Camp, Appointment
Constructors	Donor/donation/camp initialization
Array	Blood group information
ArrayList	Donors and camps
LinkedList	Donation history
HashMap	Donor ID → Donor
TreeMap	Donors sorted by blood group/name
CRUD	Donor/camp/appointment management
Searching	Search donor by ID/blood group
Sorting	Sort donors by name/donation date
Swing	Donation camp GUI

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Exception Handling	Invalid donation/appointment
Validation	Blood group, donor age, contact details

Expected Modules

- Donor Registration
- Blood Group Management
- Donation Camp Management
- Appointment Management
- Donation Recording
- Donor Search
- Donor Sorting
- Donation History
- Certificate Generation
- Donation Report

Conclusion

The project demonstrates a practical healthcare application while allowing students to implement Java OOP, collections, searching, sorting, GUI programming, validation, and exception handling.
